# Autonomous Agent Wallet & Risk Control Plane

2026 Shenzhen International FinTech Competition (FinTechathon) - Topic F

- **Theme: AI as a Financial Participant (Autonomous On-Chain Wallet)**
- **Core Thesis: Pair AI conversational planning with deterministic financial guardrails**
- **Public Repository: https://github.com/Crux-15/agent-wallet**
- **Verification: 50 Automated Pytest Unit Tests + Live Ethereum Sepolia Execution**

# 1. The Problem: Why Naive AI Crypto Wallets Fail

Unbounded LLM agency creates unacceptable financial and security exposure

- **Prompt Injection: Malicious text can trick an LLM into draining wallet funds**
- **Hallucinated Parameters: LLMs can miscalculate decimals, addresses, or gas fees**
- **No Velocity Controls: Repeated small transfers can bypass single-payment checks**
- **Solution: Strict separation between Cognitive Plane (AI) and Execution Plane (Policy + Keys)**

# 2. System Architecture: Control Plane vs. Execution Plane

Defense-in-depth across four modular software layers

- **1. UI & Supervisor Layer: Streamlit Web Control Center (app.py) & CLI (agent.py)**
- **2. Cognitive Layer (agent.py): Gemini function-calling + deterministic local fallback router**
- **3. Guard Layer (executor.py): ToolExecutor whitelist, human approval gate, JSONL audit log**
- **4. Policy & Adapter Layer: PolicyEngine (policy.py) + MockAdapter & Sepolia ChainAdapter**

# 3. Autonomous Asset Management (Task Completion - 40%)

Six structured tools give the agent complete treasury visibility and execution power

- **get_portfolio_status: Tracks total balance, 0.002 ETH reserve floor, and spendable liquidity**
- **simulate_transfer: Preflight dry-run returning 0-100 risk score, gas cost, and projected balance**
- **send_eth & batch_send_eth: Single and multi-recipient settlement with per-item policy checks**
- **get_balance & get_policy_limits: Real-time inspection of wallet state and active governance rules**

# 4. Three-Tier Governance & 7-Stage Policy Pipeline (Security - 30%)

Deterministic Decimal math and persistent 24-hour state (logs/policy_state.json)

- **Tier 0 (READ_ONLY): Blocks all outgoing transfers; allows monitoring and dry-run simulation**
- **Tier 1 (APPROVE_ALL): 100% of valid transfers pause for human operator approval**
- **Tier 2 (LIMITED_AUTO): Auto-executes <= 0.002 ETH; escalates > 0.002 ETH; denies > 0.01 ETH**
- **Hard Guards: Allowlist regex check, 0.05 ETH 24h cap, 0.002 ETH reserve floor, 50 Gwei gas ceiling**

# 5. Quantitative Risk Scoring & Preflight Simulation (Innovation - 30%)

Every transaction is scored from 0 to 100 before touching the blockchain

- **Risk Formula combines recipient trust (+60), size ratio (+40), threshold (+15), 24h cap (+35), gas (+25)**
- **Risk Bands: LOW (0-19), MEDIUM (20-39), HIGH (40-69), CRITICAL (70-100)**
- **Commit-After-Receipt: Daily spend cap updates ONLY after on-chain receipt.status == 1**
- **Resilient Routing: Local intent router guarantees zero downtime during LLM API rate limits**

# 6. Interactive Web Control Center (Streamlit Dashboard)

Real-time visual telemetry, preflight risk simulator, and human-in-the-loop approval

- **Top KPI Bar: Network mode, Total ETH, Spendable ETH above reserve, 24h cap, Live Gas (Gwei)**
- **Sidebar Governance: Instant switching between Mock/Sepolia and Permission Tiers 0, 1, and 2**
- **Supervisor Banner: One-click Approve or Reject workflow for medium-risk transfers**
- **Live Audit Trail: Real-time table rendering structured events from logs/audit.jsonl**

# 7. Real On-Chain Execution Evidence (Ethereum Sepolia Testnet)

Verified live blockchain transactions with Etherscan proof and audit logs

- **Recipient Address: 0x78964c29614bF783c3339cd7b553C8627168554E**
- **TX #1 (Auto-Approved 0.0005 ETH | Risk 1/100 LOW | 2026-10-19 22:44:00):**
- **0xd03fff8dc8ba123539f97cab7dd456c9487535c446df01674aee8cee6524a0a4**
- **TX #2 (Human-Approved 0.0022 ETH | Risk 21/100 MEDIUM | 2026-10-19 22:44:38):**
- **0xbde5dd8c3f4e0ffdd0afae4ae95ae29e58f183618a17314e07495a323ac17cc0**

# 8. Testing Rigor & Honest Security Self-Assessment

50 passing unit tests and a transparent production hardening roadmap

- **50 Pytest Unit Tests: Covers adapters, rolling 24h persistence, reserve floor, and prompt injection**
- **Tier Verification Proof: Identical 0.001 ETH request -> Denied (T0), Human Asked (T1), Sent (T2)**
- **Known Prototype Risks Documented: Local approval channel, .env key storage, local JSON state**
- **Production Roadmap: ERC-4337 Account Abstraction session keys, WebAuthn passkeys, HSM custody**

# 9. Summary: Built for Safe, Autonomous On-Chain Finance

Complete alignment with FinTechathon Topic F evaluation criteria

- **Task Completion (40%): Full portfolio telemetry, simulation, single & batch Sepolia execution**
- **Security & Compliance (30%): 3 permission tiers, 7-stage policy engine, sanitized JSONL audit log**
- **Innovation & Interaction (30%): 0-100 risk scorer, Streamlit Control Center, resilient dual router**
- **Code & Evidence Ready: https://github.com/Crux-15/agent-wallet**